

Modes of Convergence

Almost sure, in probability, L^p , and in law — how they imply each other

July 3, 2026

Outline

Definitions

The implication map

Counterexamples

Special cases

Summary

Four notions of “ $X_n \rightarrow X$ ”

Let $X_1, X_2, \dots$ and X be random variables on the same probability space $(\Omega, \mathcal{F}, P)$.

Almost surely (a.s.)

$$X_n \xrightarrow{\text{a.s.}} X \iff P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1$$

In probability

$$X_n \xrightarrow{P} X \iff \forall \varepsilon > 0, \quad P(|X_n - X| > \varepsilon) \rightarrow 0$$

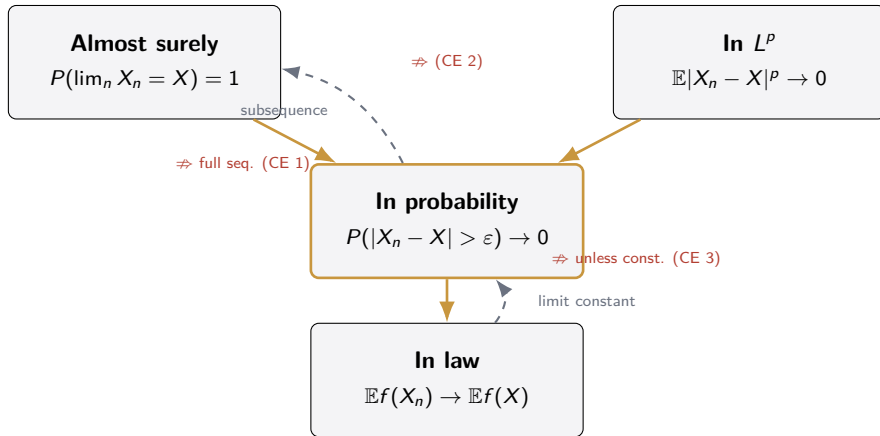
In L^p , $p \geq 1$ (p -th mean)

$$X_n \xrightarrow{L^p} X \iff \mathbb{E}|X_n - X|^p \rightarrow 0$$

In law (weakly)

$$X_n \xrightarrow{d} X \iff \mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)] \quad \forall f \text{ bounded continuous}$$

Which convergence implies which



— always true - - - true under an extra condition $\not\Rightarrow$ false in general (see counterexample)

Not drawn, since it follows transitively: $L^p \Rightarrow$ in law. Also false in general: in probability $\not\Rightarrow L^1$ (CE 4).

Counterexample 1: in probability $\nRightarrow$ almost surely

Setup. On $(\Omega, P) = ([0, 1], \text{Lebesgue})$. Write each $n \geq 1$ uniquely as $n = 2^k + j$ with $0 \leq j < 2^k$, and set

$$A_n = \left[\frac{j}{2^k}, \frac{j+1}{2^k} \right], \quad X_n = \mathbf{1}_{A_n}.$$

In probability. $P(A_n) = 2^{-k} \rightarrow 0$, so $X_n \xrightarrow{P} 0$.

Not a.s. The intervals sweep across $[0, 1]$ forever: every ω lies in infinitely many A_n and infinitely many complements, so $X_n(\omega)$ keeps oscillating between 0 and 1 — it converges at no point at all.

the typewriter sequence

Counterexample 2: almost sure $\nRightarrow L^p$

Setup. On $([0, 1], \text{Lebesgue})$, let $X_n = n \cdot \mathbf{1}_{(0, 1/n)}$.

Almost surely. For every fixed $\omega > 0$, eventually $1/n < \omega$, so $X_n(\omega) = 0$. Hence $X_n \xrightarrow{\text{a.s.}} 0$ (and therefore also $X_n \xrightarrow{P} 0$).

Not in L^p . $\mathbb{E}[X_n] = n \cdot \frac{1}{n} = 1$ for every n — the mean never moves. So $X_n \not\xrightarrow{L^p} 0$, for any $p \geq 1$.

the escaping spike — no domination, no uniform integrability

Counterexample 3: in law $\nRightarrow$ in probability

Setup. Let $X \sim N(0, 1)$ and $X_n = (-1)^n X$.

In law. Since $-X \sim N(0, 1)$ too, every X_n has exactly the law of X . So trivially $X_n \xrightarrow{d} X$.

Not in probability. For odd n , $|X_n - X| = 2|X|$, which is not small. $P(|X_n - X| > \varepsilon)$ does not tend to 0: convergence in law only pins down marginal distributions, never the coupling between X_n and X .

same law, different variable

Counterexample 4: in probability $\not\Rightarrow L^1$

Setup. Reuse the typewriter sets $A_n = [\frac{j}{2^k}, \frac{j+1}{2^k}]$ from CE 1 ($n = 2^k + j$), but raise the spike height with the level:

$$X_n = 4^k \cdot \mathbf{1}_{A_n}.$$

In probability. As before, $P(A_n) = 2^{-k} \rightarrow 0$, so $X_n \xrightarrow{P} 0$.

Not in L^1 . $\mathbb{E}[X_n] = 4^k \cdot 2^{-k} = 2^k \rightarrow \infty$: the mean doesn't just fail to vanish, it diverges. Convergence in probability puts no control at all on L^1 behaviour — only on how *often* a deviation happens, never on its *size*.

the typewriter sequence, with the volume turned up

Three repairs that restore a missing arrow

Subsequence rescue (F. Riesz)

$$X_n \xrightarrow{P} X \Rightarrow \exists \text{ a subsequence } X_{n_k} \xrightarrow{\text{a.s.}} X.$$

Constant-limit rescue

$$X_n \xrightarrow{d} c, c \text{ constant} \Rightarrow X_n \xrightarrow{P} c.$$

Uniform-integrability rescue

$$X_n \xrightarrow{\text{a.s.}} X \text{ (or } \xrightarrow{P} \text{)} \text{ and } \{|X_n|^p\} \text{ uniformly integrable} \Rightarrow X_n \xrightarrow{L^p} X \text{ (dominated convergence).}$$

Every pairwise relation, at a glance

if $X_n \rightarrow X$ in...	a.s.	in P	L^p	in law
a.s.	—	✓	$\nRightarrow^\dagger$	✓
in P	$\nRightarrow^*$	—	$\nRightarrow^\dagger$	✓
L^p	$\nRightarrow^*$	✓	—	✓
in law	$\nRightarrow$	$\nRightarrow^\dagger$	$\nRightarrow$	—

* holds for a subsequence only (F. Riesz)

† holds if the limit is almost-surely constant

‡ holds under an added uniform-integrability / domination condition

Almost-sure and L^p convergence are the strong notions; both flow through convergence in probability into the weakest one, convergence in law — and none of the arrows run backward without borrowing an extra condition.